KAIZEN MATHS

Lesson starter

Number Bases and Number Sets

Base Operations

Name _____

S1

Calculate $132_8 + 2_8$. Give your answer in base 8.

S2

Calculate $101_3 - 10_3$. Give your answer in base 3.

S3

Calculate $55_6 \times 4_6$. Give your answer in base 6.

S4

Find the base k if $55_k = 35_{10}$.

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NUM-5141-L3 · S

Show your working

KAIZEN MATHS

Exit ticket

Number Bases and Number Sets

Base Operations

Name _____

E1

Find the digit x if x1 5 = 6 10 .

E2

Calculate 67 8 + 41 8 . Give your answer in base 8.

E3

Calculate 102 3 - 100 3 . Give your answer in base 3.

KAIZEN MATHS

Exit ticket

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NUM-5141-L3 · E

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NUM-5141-L3 · E

Show your working

Teacher answers & working

Number Bases and Number Sets · Base Operations

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S1

Calculate $132_8 + 2_8$. Give your answer in base 8.

Answer:

134_8

Working:

Convert each number to denary first.

$132_8 = 90_{10}$

$2_8 = 2_{10}$

$90 + 2 = 92$

Convert 92_{10} back to base 8: 134_8.

S2

Calculate $101_3 - 10_3$. Give your answer in base 3.

Answer:

21_3

Working:

Convert each number to denary first.

$101_3 = 10_{10}$

$10_3 = 3_{10}$

$10 - 3 = 7$

Convert 7_{10} back to base 3: 21_3.

S3

Calculate $55_6 \times 4_6$. Give your answer in base 6.

Answer:

352_6

Working:

Convert each number to denary first.

$55_6 = 35_{10}$

$4_6 = 4_{10}$

$35 \times 4 = 140$

Convert 140_{10} back to base 6: 352_6.

Teacher answers & working

Number Bases and Number Sets · Base Operations

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S4

Find the base k if $55_k = 35_{10}$.

Answer:

6

Working:

Expand using place value in base k.

$55_k = 5k + 5$

$5k + 5 = 35$

$5k = 30$

$k = 6$

Check: every digit must be less than the base. The largest digit is 5, and $5 < 6$.

E1

Find the digit x if $x1_5 = 6_{10}$.

Answer:

1

Working:

Expand using place value in base 5.

$x1_5 = 5x + 1$

$5x + 1 = 6$

$5x = 5$

$x = 1$

Check: 1 is a valid base-5 digit because $1 < 5$.

E2

Calculate $67_8 + 41_8$. Give your answer in base 8.

Answer:

130_8

Working:

Convert each number to denary first.

$67_8 = 55_{10}$

$41_8 = 33_{10}$

$55 + 33 = 88$

Convert 88_{10} back to base 8: 130_8.

Teacher answers & working

Number Bases and Number Sets · Base Operations

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E3

Calculate 102 3 - 100 3 . Give your answer in base 3.

Answer:

2_3

Working:

Convert each number to denary first.

$102_3 = 11_{10}$

$100_3 = 9_{10}$

$11 - 9 = 2$

Convert 2_{10} back to base 3: 2_3.